FLAGSHIP SOLUTION SCALING TRACK PROFILE

Scaling Delivery Systems for Carp, Tilapia, and Small Indigenous Species (SIS) Genetics and Input Services in Bangladesh

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 2 · Deliver Inputs and Services at Scale through Inclusive Market Systems

Flagship 2.3 · Genetics, Inputs & Services for Animals & Aquaculture

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

  Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

  Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

  Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

  Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

  Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

  Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Aquaculture is a cornerstone of Bangladesh's rural economy, contributing 59% of national fish production and supporting the diets and incomes of over 18 million people. However, its growth is constrained by a fragmented and low-trust seed and input delivery system. Farmers frequently receive poor-quality seed due to weak broodstock management, limited certification, and dispersed nursery networks, resulting in lower survival rates, slower growth, and reduced profitability. Despite the development of improved fish strains through CGIAR research, these advances rarely reach the last mile, particularly for women, youth, and marginalized groups in climate-vulnerable areas.

This Solution Track aims to transform aquaculture delivery by scaling a decentralized and quality-assured system centered on an innovative partnership model called SeedCompacts, to expand reliable access to improved carp, tilapia, and Small Indigenous Species (SIS) genetics and input services. The primary geographical focus is Bangladesh, with targeted South–South learning exchanges planned with Kenya. The core innovation is not merely the improved genetics themselves, but the bundled delivery mechanism that links licensed hatcheries, nurseries, and seed traders into accountable networks.

This system is reinforced by advisory and coordination support through the OneFish platform, phased development of digital traceability systems to strengthen seed Quality Assurance (QA), and collaboration with NGOs that implement nutrition programs, funded through various sources (e.g., government and donors), to train their Community Health Workers (CHWs) to include fish-related nutrition messages in their routine outreach activities. Targeted system strengthening interventions, including diagnostics, practical Standard Operating Procedures (SOPs) for hatcheries and nurseries, access to finance mechanisms, and capacity development, support consistent and scalable performance.

The scaling pathway is grounded in robust evidence, including on-farm trials showing genetically improved rohu delivering 33 to 37% higher harvest weights and pilot initiatives where decentralized extension reached over 10,000 farmers, generating an additional USD 1.27 million in production value. The track is led by WorldFish in close collaboration with the International Food Policy Research Institute (IFPRI). The track will engage government partners such as the Department of Fisheries (DoF) and Bangladesh Fisheries Research Institute (BFRI), private sector actors, and financial partners including the Palli Karma-Sahayak Foundation (PKSF), leveraging their aquaculture-funded networks to build staff capacity for disseminating technical knowledge to beneficiaries and ensuring their seamless integration into SeedCompact networks.

2. Innovation Bundle and Scaling Ambition

2.1 What is Being Scaled

- **SeedCompact Model:** A decentralized delivery network organizing licensed hatcheries, nurseries, and seed traders into accountable groups to supply verified, high-quality seed.
- **Improved Genetics**: Dissemination of genetically improved carp (e.g., G3 rohu), tilapia, and SIS broodstock developed in coordination with the Sustainable Animal and Aquatic Foods (SAAF) program.
- **Advisory and Digital Support:** Deployment of the OneFish platform to provide farm-specific advisory, service coordination, and market linkages to farmers and delivery actors.
- **Digital Traceability Systems:** Development and piloting of traceability solutions, initially at the hatchery level, to enable tracking of seed origin and key quality attributes, with scope for phased expansion across the broader seed delivery network to strengthen system transparency and trust.
- **Nutrition Messaging and Outreach:** Dissemination of fish-based nutrition messages through NGO-led CHWs, supporting improved dietary awareness and increased household consumption of nutrient-rich fish.

- **System Strengthening Functions:** Targeted interventions including diagnostic assessment of system bottlenecks, development of practical SOPs for hatchery and nursery, facilitation of access to finance, and capacity development to ensure consistent and scalable system performance.

2.2 Evidence of Impact

- **Genetic Performance:** On-farm trials of improved rohu (G3) demonstrate 33–37% higher harvest weights and 24% higher profitability compared to local strains, with improved survival rates (Hamilton et al., 2022; Ali et al., 2025).
- **Pilot Effectiveness:** AMD pilots (2023–24) with 101 nurseries and 30 input dealers reached 10,234 farmers across 1,907 ha, generating approximately 610 tons of additional rohu production valued at USD 1.27 million (Halder and Biswas, 2024).
- **Extension and Nutrition:** Decentralized extension by local service providers such as nurseries and feed dealers increased productivity of farmers by up to 25% and enhanced household food security (Dompreh et al., 2024; Obi et al., 2025). NGO-led CHWs reached over 52,000 households with fish-specific nutrition messages, improving women's dietary diversity.
- **Private Sector Engagement:** Corporate Social Responsibility (CSR) financing from a private bank mobilized USD 170,000 for G3 rohu distribution, demonstrating private-sector willingness to co-invest.

2.3 Scaling Targets

Farmers reached	~130,000 smallholder farmers	400,000 farmers (≥20% women and youth)
Area under improved practices	~27,000 ha	~84,000 ha
Jobs supported	Baseline established	11,000 FTE jobs supported and/or enhanced
Consumer access	Baseline established	100,000 consumers with improved access to aquatic foods via increased market availability.
Hatcheries engaged in SeedCompact networks	15 hatcheries	30 hatcheries
Nurseries engaged in SeedCompact networks	600 nurseries	1,800 nurseries
Seed traders engaged in SeedCompact networks	100 seed traders	300 seed traders

3. Scaling Pathway and Theory of Change

3.1 Pathway Strategy

The scaling pathway is a hybrid model that leverages public, private, and community channels to achieve systemic change. It is designed to be partner-led, avoiding the creation of parallel structures.

First, the pathway creates market pull by engaging institutional partners such as DoF to embed quality assurance in national procedures, and private sector actors to create predictable demand for verified inputs.

Second, it strengthens last-mile delivery through SeedCompacts, which formalize and organize licensed hatchery, nursery, and seed trader networks. These compacts are supported by capacity building, practical SOPs, advisory support through digital platforms (OneFish App), and phased development of digital traceability systems, thereby building trust and reducing quality variability.

Third, the pathway ensures inclusivity and sustainability by integrating nutrition outreach through NGO-led CHWs, facilitating access to finance through partnerships with financial institutions and intermediaries, and generating operational evidence through the Monitoring, Evaluation, Learning, and Impact Assessment (MELIA) to inform policy dialogue.

In parallel, targeted system strengthening interventions—including diagnostic assessment of system bottlenecks, development of practical SOPs for hatchery and nursery, and capacity development of delivery actors—are implemented to ensure consistent system performance and scalability.

The ultimate goal is for DoF to institutionalize the decentralized delivery model and QA standards within national aquaculture extension and subsidy systems.

3.2 Theory of Change

If smallholder farmers in Bangladesh gain reliable access to a bundle of improved carp, tilapia, and SIS genetics, advisory services, and quality-assured inputs through decentralized and accountable SeedCompact networks, and if key market and coordination failures—such as weak traceability, fragmented supply chains, and limited access to working capital—are addressed through partnership governance, phased deployment of traceability systems, strengthened coordination mechanisms, and improved access to financial and advisory services, then production risk for farmers will be reduced, leading to higher productivity, profitability, and adoption of good practices, leading to increased incomes and enhanced employment opportunities along the value chain, and improved consumption of nutritious fish, ultimately contributing to a more resilient, inclusive, and sustainable aquaculture system in Bangladesh, with delivery models and quality standards institutionalized by national government.

3.3 Key Activities Funded by S4I

- **Partnership Formation:** Contracting and formalizing SeedCompacts with licensed hatcheries, nurseries, seed traders, and relevant service and digital partners.
- **Capacity Building:** Integrated training on good aquaculture practices, advisory integration, monitored delivery, compliance with SOPs, and system-level coordination.
- **Governance and Standards Alignment:** Multi-stakeholder consultations with DoF and BFRI to align certification standards and practical SOPs, traceability principles, and governance roles for national systems.
- **Digital Traceability Development and Piloting:** Design and piloting of traceability solutions, initially at the hatchery level, with scope for phased expansion across the seed delivery network, including nurseries and seed traders.
- **Performance Monitoring and Learning:** Baseline assessments, performance monitoring (including field tracking and telephonic follow-ups), and outcome documentation to track delivery performance, system change, and inclusion outcomes, and to generate evidence for adaptive management and policy engagement.
- **Knowledge Synthesis and Exchange:** Supporting learning events for institutional uptake and cross-country adaptation (e.g., with Kenya).

4. Partners and Roles

Demand Partners	**Department of Fisheries (DoF):** National agency. **Bangladesh Fisheries Research Institute (BFRI):** National research body. PKSF + MFI/NGO Networks: Umbrella organization for microfinance. **Private Sector:** Hatcheries, nurseries, seed traders, input dealers, processors, and market actors.	**DoF:** Creates institutional pull for quality seed through inspection, extension, and plans to embed QA in national procedures. **BFRI:** Provides technical legitimacy by co-developing broodstock, performance, and biosecurity standards. **PKSF/MFIs:** Aggregates farmer demand and enables access to working capital for producers and service providers. **Private Sector:** Creates predictable demand for quality-assured seed and inputs and strengthens market linkages by connecting producers to input and output markets.
Scaling Partners	Licensed Hatcheries (**30 by 2028**), Nurseries (**1,800 by 2028**), Seed Traders (**300 by 2028),** and Local Service Providers: Core of the SeedCompact model. PKSF-linked NGOs: Implementing partners for community outreach. DoF and WorldFish Delivery Platforms: Selected ongoing projects. ARITS Limited: Digital partner.	**Hatcheries/Nurseries/Seed Traders/LSPs:** Form the accountable SeedCompact networks that supply improved and quality-assured seed, provide advisory support, and enable monitored last-mile delivery. **NGOs:** Train and deploy CHWs to integrate fish-specific nutrition advice into ongoing programs, linking adoption to dietary outcomes. **DoF/WorldFish Platforms**: Embed SeedCompact approaches into ongoing programs to expand reach, build frontline capacity, and generate evidence. **ARITS:** Supports the OneFish platform for advisory, service coordination, and market linkages; separate traceability solutions are developed and piloted to strengthen system transparency and quality assurance.

Funding Model

S4I provides catalytic investments to organize, test, and strengthen the decentralized delivery system (e.g., partnership formation, capacity building, MELIA), while core inputs (seed, feed) and services are financed through private-sector sales, public-sector programs, and facilitated access to finance through microfinance and other financial partners.

5. Enabling Environment and Risks

5.1 Opportunities

- **Strong Government Commitment:** DoF and BFRI have demonstrated strong interest in strengthening QA, practical SOPs, and system-level improvements within the existing regulatory framework of the aquaculture sector.
- **High Partner Readiness:** A high proportion of hatcheries are licensed, providing a strong regulatory foundation for engaging seed system actors through organized delivery networks.

- **Proven Economic Incentives:** Improved strains command a 10–20% price premium at both the hatchery (spawn) and nursery (fingerling) levels, creating a strong commercial logic for actors to maintain quality and participate in organized delivery networks and QA systems.
- **Leverage from Complementary Investments:** The track is closely aligned with bilateral projects (e.g., Sustho Sagor project in Bangladesh; Climate-Resilient Aquaculture Systems for Africa (CASA) project in Kenya) and the core WorldFish Genetics Program, ensuring a pipeline of improved genetics and additional outreach channels.
- **Strong Evidence Base for Scaling:** The track builds on robust evidence from on-farm genetic performance trials, decentralized extension pilots, and previous network-based delivery initiatives, providing a strong foundation for scaling with confidence.

5.2 Challenges and Mitigation

- **Governance and Quality Assurance Risks:** Weak coordination among seed system actors could undermine delivery quality.
- **Mitigation:** Formalize SeedCompacts with clear bylaws, roles, and traceability systems. Introduce phased digital traceability solutions, beginning at the hatchery level, to strengthen transparency and trust. Co-develop practical SOPs with DoF and BFRI to ensure national alignment.
- **Financial Risks:** Limited capital for hatchery upgrades and nursery expansion may slow adoption.
- **Mitigation:** Engage PKSF and MFIs and other relevant financial actors to expand credit lines for working capital. Use S4I catalytic funds for training and QA, not direct working capital, to de-risk early investment.
- **Policy and Institutional Risks:** The time required for formal recognition or institutional uptake of the decentralized delivery model by DoF may affect the pace of scaling and integration into national systems.
- **Mitigation:** Maintain continuous engagement with DoF through technical dialogue and alignment with existing regulatory frameworks, while generating practical implementation evidence through phased piloting to inform and support gradual institutional uptake.
- **Equity and Inclusion Risks:** Women, youth, and marginalized groups may be excluded from benefits.
- **Mitigation:** Set and monitor explicit targets (≥20% women and youth). Integrate gender-sensitive nutrition messaging via CHWs. Prioritize outreach in climate-vulnerable areas through delivery actors and partner networks
- **Climate Risks:** Floods, droughts, and salinity can disrupt seed supply chains.
- **Mitigation:** Diversify production and distribution points where feasible, strengthen coordination across delivery actors, and promote climate-resilient broodstock management practices.
- **Implementation Considerations:** The effectiveness of delivery networks depends on timely development and rollout of enabling functions such as SOPs, access to finance mechanisms, and traceability systems.
- **Mitigation:** Sequence system strengthening activities through diagnostics, phased piloting, and targeted capacity development, ensuring that delivery expansion is aligned with the availability of practical tools, financial access mechanisms, and coordination support.

6. Learning and Adaptation

6.1 Monitoring Focus

The MELIA framework will track progress and generate evidence for adaptive management and policy dialogue, focusing on:

- **Process Outcomes:** Functionality of SeedCompact networks (e.g., adherence to bylaws, quality of partnerships), reach and quality of capacity-building activities, and effectiveness of coordination among delivery actors.
- **Behavior Change:** Adoption of good practices by hatcheries/nurseries (e.g., maintaining separate ponds for improved strains, progressive use of traceability systems where applicable; integration of QA standards by DoF; use of OneFish by farmers and service providers.
- **System Outcomes:** Seed quality and progressive traceability performance along the value chain; reduction in transaction costs; evidence of private-sector co-investment; changes in the enabling environment (e.g., adoption or reference of standards and practices in institutional processes).
- **Inclusion and Impact:** Disaggregated data on farmer reach (gender, youth), job support and/or enhance, adoption rates, productivity gains, and nutrition outcomes.
- **Innovation Readiness and Use:** Application of the IPSR innovation packaging process to assess readiness, use, and key bottlenecks, generating structured evidence to inform adaptive management and scaling decisions.

6.2 Adaptive Management

- **Structured Review and Check-ins:** Regular engagement with the S4I team to review progress against outcome targets, identify emerging challenges, and adapt implementation approaches based on evidence.
- **Data-Driven Feedback Loops:** Use of monitoring data, field tracking, and digital reporting tools to generate timely feedback for delivery actors and implementing teams, supporting continuous improvement.
- **Peer Learning:** Structured cross-country learning exchanges (e.g., with the Kenya team) and participation in S4I-wide learning events to share insights and refine strategies.
- **Periodic Review and Stage-Gating Support:** Contribution of monitoring and IPSR-generated evidence to S4I portfolio reviews and stage-gating processes, supporting assessment of progress, identification of bottlenecks, and refinement of scaling strategies.

7. Next Steps and Timeline (2026)

Q1 (Jan-Mar)	**Partnership Formation and Setup:** Initiate and advance the selection and engagement of hatcheries, nurseries and seed traders (with selection extending into early Q2). Begin multi-stakeholder consultations with the DoF, BFRI, and private sector partners to conduct a preliminary assessment of existing seed traceability practices at hatchery level, identify requirements for practical SOPs, and ensure alignment with system needs. Initiate discussions on baseline survey design and tool development.
Q2 (Apr-Jun)	**Partnership Finalization and Capacity Building:** Complete the selection and formalization of SeedCompact partners. Organize engagement events bringing together hatcheries, nurseries, and fish seed traders to build linkages and raise initial awareness on good aquaculture practices, practical SOPs, digital systems, and advisory integration. Initiate preliminary work to identify and engage a digital partner for traceability system development, in consultation with the DoF and BFRI. Organize the IPSR workshop and finalize tools and survey design for the baseline assessment of aquaculture value chain actors.
Q3 (Jul-Sep)	**Operational Piloting and System Diagnostics:** SeedCompacts become operational, delivering seed and advisory services during the main farming season. Complete baseline data collection and initiate early performance monitoring. Undertake system diagnostics to assess bottlenecks in seed quality, coordination, and service delivery, and initiate the design of digital traceability solutions (with initial focus at hatchery level). Begin drafting practical SOPs based on diagnostic findings and stakeholder consultations. Conduct structured learning workshops with partners to review early implementation and refine approaches. Facilitate South–South exchange and learning activities (e.g., with Kenya) to share experiences and inform adaptive improvements.
Q4 (Oct-Dec)	**Traceability Piloting, Learning and Planning:** Initiate piloting of digital traceability solutions. Conduct performance monitoring and synthesize operational evidence. Begin capacity development and training for hatcheries, nurseries, and seed traders based on finalized SOPs to strengthen compliance and delivery quality. Support policy dialogue with the DoF through evidence sharing. Refine tools and approaches based on learning and implementation feedback. Prepare for phased expansion in 2027.

Communications and Knowledge: Key products will include evidence briefs on SeedCompact performance, system-level learning, and traceability development; presentations at national aquaculture workshops; and contributions to CGIAR and S4I knowledge-sharing platforms. Partner experiences and implementation insights will be documented to support learning and wider uptake.

8. Contact

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